Problem 6: Max of All Subarrays of Size 'k'

Problem Statement

Given an integer array and an integer `k`, design an algorithm to find the maximum for each and every contiguous subarray of size `k`.

Examples:

1. **Input:** array = `[1, 2, 3, 1, 4, 5, 2, 3, 6]`, k = 3
   Output: `[3, 3, 4, 5, 5, 5, 6]`
   Description: Here, subarray `1,2,3` has maximum 3, `2,3,1` has maximum 3, `3,1,4` has maximum 4, `1,4,5` has maximum 5, `4,5,2` has maximum 5, `5,2,3` has maximum 5, and `2,3,6` has maximum 6.

2. **Input:** array = `[8, 5, 10, 7, 9, 4, 15, 12, 90, 13]`, k = 4
   Output: `[10, 10, 10, 15, 15, 90, 90]`
   Description: Here, the maximum of each subarray of size 4 are 10, 10, 10, 15, 15, 90, 90 respectively.

3. **Input:** array = `[1,2,3,4,5]`, k = 3
   Output: `[3, 4, 5]`
   Description: Here, the maximum of each subarray of size 3 are 3, 4, 5 respectively.

Try it yourself

Try solving this question here:

Java

```java
import java.util.ArrayList;
import java.util.Deque;
import java.util.LinkedList;

public class Solution {
    public List<Integer> printMax(int[] arr, int k) {
        List<Integer> result = new ArrayList<Integer>();
        // ToDo: Write Your Code Here.
        return result;
    }
}
```

Test Case | Run Code Result | Stdout

```
[6, 7, 8, 9, 10]
2
```

Use Example Testcases | Run | Submit

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Solution: Max Of All Subarrays Of Size 'K'

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